

Subject Description Form

Subject Code	ITC3025T
Subject Title	Coloration and Finishing Technology
Credit Value	3 credits
Level	3
Pre-requisite/ <Co-requisite> / (Exclusion)	ITC2007T Textile Studies III
Objectives	This subject aims to provide students with in depth knowledge in textile wetting processing and prepare students with extended knowledge and understanding of principles and technology of coloration and finishing with special emphasis on the chemistry and mechanism of the processes.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand and describe the capabilities and limitations of the various pre-treatment, dyeing and finishing processes in specific applications. - Apply their knowledge of chemical mechanisms and technologies related to textile preparation, dyeing and finishing (i.e. wet processing), to select the correct machinery, processes, processing conditions and technologies to achieve the specific effects required for different end-uses and to minimize and solve technical problems, including environmental ones. - Describe and compare the influence of varying processing conditions on the properties of the finished products.
Subject Synopsis/ Indicative Syllabus	(I) Coloration The chemical principle of coloration/dyeing. Dye classifications and dye properties. Dyes and dyeing for popular textile fibers such as cotton/cellulosic, protein, and polyester Fibers. Factors affecting dyeing, process parameters, and control. (II) Finishing Mechanical finishing Mercerization, finishing with enzyme, softening finishing Easy-care and durable press finishing of cellulose, anti pilling finishes Repellent finishes, soil release finishes Flame-retardant finishes, antistatic finishes Antimicrobial finishes, ultraviolet protection finishes Novel finishes
Teaching/Learning Methodology	The principle and technology of typical dyeing and finishing processes are explained and discussed in lectures with emphasis on the methods of application and the chemistry, mechanism, and control of the processes. Lab experiments are designed to provide students with hands-on opportunities to deepen their understanding on the concepts taught in lectures and to enhance their practical skills of using different techniques.

Assessment Methods in Alignment with Intended Learning Outcomes								
	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c			
	1. Assignments	25%	✓	✓	✓			
	2. Lab reports	25%	✓	✓	✓			
	3. Exam.	50%	✓	✓	✓			
	Total		100%					
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:								
1. Assignment: There are some questions listed in teaching notes for each lesson. Students will be given two assignments to answer the assigned questions by written based on the theory they have learnt in the class. 2. Lab report: After each lab experiment, students are required to write a lab report which should involve the parts of introduction, experimental, results and discussion, conclusion, and references related to the experiment. 3. Final examination: Students will need to answer questions about the fundamental concepts of coloration and finishing technologies and their applications.								
Student Study Effort Expected	Class contact:							
	▪ Lecture		26 Hrs.					
	▪ Studio		12 Hrs.					
	Other student study effort:							
	▪ Assignments		62 Hrs.					
	▪ Quiz and Exam.		4 Hrs.					
	Total student study effort		104 Hrs.					
Reading List and References	Books							
	Arthur D. Broadbent (2001). <i>Best Principles of Textile Coloration</i> . SDC. Perkins, W.S. (1996). <i>Textile Coloration and Finishing</i> . Durham. N.C., Carolina Academic Press. Broadbent, A.D. (2001). <i>Basic Principles of Textile Coloration</i> . The Society of Dyers and Colourists. W. D. Schindle and P.J. Hauser, <i>Chemical finishing of textiles</i> , The Textile Institute Edited by Derek Heywood, <i>Textile finishing</i> , 2003, Society of Dyers and Colourists							